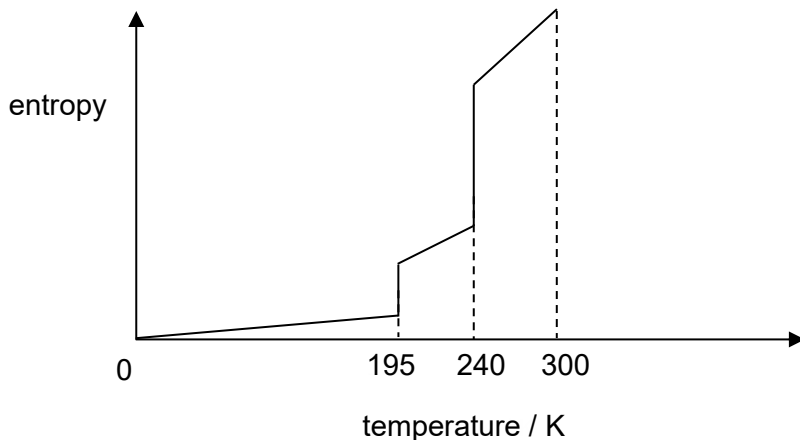


- 11 The melting point of ammonia is 195 K and the boiling point of ammonia is 240 K.

The graph below shows how the entropy of NH_3 changes between 0 K and 300 K.



Which statements explain the shape of the graph?

- 1 Between 0 K and 195 K, entropy is low as the NH_3 molecules are held in their fixed positions in the solid state.
- 2 Between 195 K and 240 K, there is an increase in the number of ways to distribute energy among NH_3 molecules.
- 3 At 240 K, there is an increase in the number of NH_3 molecules and hence number of ways to arrange NH_3 molecules.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2, and 3